Technical Document #112: Retrieval Augmented Generation - Comprehensive Overvi

Technical Document #112: Retrieval Augmented Generation - Comprehensive Overview

Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Query decomposition breaks complex queries into simpler sub-queries. It improves retrieval quality for multi-faceted questions.

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. It grounds LLM responses in factual, retrieved documents.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Key Takeaways:

- Hybrid search combines keyword-based and semantic search.
- Query decomposition breaks complex queries into simpler sub-queries.
- Vector databases store dense embeddings for semantic search.